

3. Row 2 Column 1 = Row 3 Column 1 = 0
 Row 3 Column 2 = 0

∴ The condition that for each column having a leading entry, all positions below the leading entry position must contain only 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0

∴ The condition that for each column having a leading entry, all positions above the leading entry position must contain only 0 is True.

So, the matrix $\begin{bmatrix} 1 & 0 & -4/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is in reduced row echelon form Π
 Column 3 has no pivots.

∴ The variable x_3 is free.

$$\begin{cases} x_1 - \frac{4}{3}x_3 = 0 \\ x_2 = 0 \\ 0 = 0 \end{cases}$$

are the corresponding equations.

∴ The solution set is

$$x_1 = \left(\frac{4}{3}\right)x_3, x_2 = 0, x_3 \text{ is free. } \square$$

A single linear equation can be treated as a very simple system "system"

Example 2:

A single linear equation can be treated as a very simple system "system"

Solution:

Describe all solutions of the homogeneous system

$$10x_1 - 3x_2 - 2x_3 = 0 \quad (1)$$

$$x_1 \begin{bmatrix} 10 \end{bmatrix} + x_2 \begin{bmatrix} -3 \end{bmatrix} + x_3 \begin{bmatrix} -2 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{bmatrix} 10 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

By defn. of $Ax=b$, $\begin{bmatrix} 10 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$

The matrix equation is now of the form $Ax=0$. The trivial solution is, $x_1=x_2=x_3=0$.

To verify whether the equation has a nontrivial solution, we must verify the existence of at least one free variable.

By defn, a free variable of a system of equations corresponds to a column of the coefficient matrix having no pivots.